

MATH 162, Winter 2026
SCRIPT 8: Intervals

Now that we have constructed $\mathbb{R}$ and proved the fundamental facts about it, we will work with the real numbers $\mathbb{R}$ instead of an arbitrary continuum C . We will leave behind Dedekind cuts and think of elements of $\mathbb{R}$ as numbers. Accordingly, from now on we will use lower-case letters like x for real numbers and will write $+$ and $\cdot$ for $\oplus$ and $\otimes$. We will also now use the standard notation (a, b) for the region $\underline{ab} = \{x \in \mathbb{R} \mid a < x < b\}$. Even though the notation is the same, this is *not* the same object as the ordered pair (a, b) .

More generally, we adopt the following standard notation:

$$\begin{aligned}
 (a, b) &= \{x \in \mathbb{R} \mid a < x < b\} \\
 [a, b) &= \{x \in \mathbb{R} \mid a \leq x < b\} \\
 (a, b] &= \{x \in \mathbb{R} \mid a < x \leq b\} \\
 [a, b] &= \{x \in \mathbb{R} \mid a \leq x \leq b\} \\
 (a, \infty) &= \{x \in \mathbb{R} \mid a < x\} \\
 [a, \infty) &= \{x \in \mathbb{R} \mid a \leq x\} \\
 (-\infty, b) &= \{x \in \mathbb{R} \mid x < b\} \\
 (-\infty, b] &= \{x \in \mathbb{R} \mid x \leq b\}.
 \end{aligned} \tag{8.1}$$

Notice that if $a = b$, then (a, b) , $[a, b)$ and $(a, b]$ are all empty, but $[a, b] = \{a\}$. We call $[a, a] = \{a\}$ a “degenerate” interval.

Exercise 8.1. Identify the sets in (8.1) that are open/closed/neither. (Don’t forget to justify your answer but you should try to be as efficient as possible in so doing.)

Exercise 8.1.

(a, b) This set is open and not closed. To see that it is open, let $p \in (a, b)$. By choosing $\delta = \min(p - a, b - p)$, the neighborhood $(p - \delta, p + \delta)$ is contained within (a, b) . It is not closed because the points a and b are limit points of the set, yet they are not contained within the set itself.

$[a, b)$ This set is neither open nor closed. It is not open because of the endpoint a where any neighborhood $(a - \delta, a + \delta)$ will contain values less than a , which are not in the set. It is not closed because b is a limit point of the set, but $b \notin [a, b)$. Since it fails to satisfy the requirement for either property, it is neither.

$(a, b]$ This set is neither open nor closed. It fails to be open because for the point $b \in (a, b]$, any δ -neighborhood will contain points greater than b which fall outside the set. It fails to be closed because the limit point a is not included in the set.

$[a, b]$ This set is closed and not open. It is closed because it contains both of its limit points, a and b . However, it is not open because for the endpoints a and b , there is no $\delta > 0$ such that the neighborhoods $(a - \delta, a + \delta)$ or $(b - \delta, b + \delta)$ are entirely contained within $[a, b]$.

(a, ∞) This set is open and not closed. For any $p \in (a, \infty)$, we can set $\delta = p - a$. The neighborhood $(p - \delta, p + \delta)$ remains within (a, ∞) because $p - \delta = a$, so all points are greater than a . It is not closed because a is a limit point that is not in the set.

$[a, \infty)$ This set is closed and not open. It is closed because its only finite limit point is a , which is included in the set. It is not open because any neighborhood around a will contain points smaller than a , which are not elements of the set.

$(-\infty, b)$ This set is open and not closed. For any p in the set, we can choose $\delta = b - p$. The resulting neighborhood $(p - \delta, p + \delta)$ stays below b , satisfying the definition of an open set. It is not closed because the limit point b is excluded.

$(-\infty, b]$ This set is closed and not open. It contains its only finite limit point b , making it closed. However, at the point b , any open neighborhood $(b - \delta, b + \delta)$ will contain points strictly greater than b , which are not in the set, thus it is not open. $\square$

Definition 8.2. A nonempty set $I \subset \mathbb{R}$ is an *interval* if, for all $x, y \in I$, with $x < y$, $[x, y] \subset I$.

Lemma 8.3. A nonempty proper set $I \subset \mathbb{R}$ is an interval if, and only if, it takes one of the eight forms in (8.1) above.

Proof of Lemma 8.3. We want to show that a nonempty proper set $I \subset \mathbb{R}$ is an interval if, and only if, it takes one of the eight forms listed in the definition.

$(\Leftarrow)$ Firstly, we show that if I follows one of the eight forms, then I is an interval. Let $a, b \in \mathbb{R}$ and suppose I takes the form (a, b) , which is the set $\{x \in \mathbb{R} \mid a < x < b\}$. To show I is an interval, let $c, d \in I$ such that $c < d$. By definition of the set, we have $a < c$ and $d < b$. Now, let z be any point such that $c \leq z \leq d$. By the transitivity of the order in $\mathbb{R}$, we see that $a < c \leq z \leq d < b$, which implies $a < z < b$. Thus, $z \in I$, so $[c, d] \subset I$. Furthermore, this argument follows similarly for the other seven forms. Therefore, I is an interval.

$(\Rightarrow)$ Now, we show that if I is an interval, then it must take one of the eight forms. Let $a = \inf I$ and $b = \sup I$, where we allow a and b to be $\pm\infty$ if the set is unbounded. By the definition of infimum and supremum, every element $x \in I$ must satisfy $a \leq x \leq b$. Furthermore, for any value z such that $a < z < b$, the properties of the infimum and supremum guarantee there exist elements $c, d \in I$ such that $a \leq c < z$ and $z < d \leq b$. Since I is an interval and $c < z < d$, it must be that $z \in I$.

This shows that I contains every point strictly between its infimum and supremum. The only remaining variation is whether the endpoints a and b themselves are members of I . Depending on whether $a \in I$ or $a \notin I$, and whether $b \in I$ or $b \notin I$, I will result in one of the eight specific forms. Therefore, any interval I must take one of the eight forms in **Definition 8.1**. $\square$

It can be useful to express bounded intervals symmetrically about their center. To do this it is natural to introduce *absolute value*.

Definition 8.4. The *absolute value* of a real number x is the non-negative number $|x|$ defined by

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

Exercise 8.5. Show that $|x| = |-x|$, for $x \in \mathbb{R}$. (Note that this also means that $|x - y| = |y - x|$ for any $x, y \in \mathbb{R}$.)

Exercise 8.5. We want to prove that $\forall x \in \mathbb{R}, |x| = |-x|$. We will prove this through two cases:

Case 1: $x \geq 0$. Suppose that $x > 0$. In this case, by **Definition 8.4**, $|x| = x$. Then, as $x > 0$, we may say that $-x < 0$ and therefore, by **Definition 8.4**, we can say that $|-x| = x$. As we have shown that $|x| = x$ and $|-x| = x$ this works to show that $|x| = |-x|$ when $x \geq 0$.

Case 2: $x < 0$. : This argument is an analogous proof to the first case.

This suffices to prove that $|x| = |-x|$. □

Definition 8.6. The *distance* between $x \in \mathbb{R}$ and $y \in \mathbb{R}$ is defined to be $|x - y|$.

Remark 8.7. It follows from Definition 8.6 that $|x|$ is the distance between x and 0.

Exercise 8.8. Prove that, if x, y are real numbers then

a) $|xy| = |x| \cdot |y|$

b) $x \leq |x|$ and $-x \leq |x|$.

Exercise 8.8, Part (a). We want to show that $|xy| = |x| \cdot |y|$. We will do this through the usage of four cases without the loss of generality.

Case 1: $x = 0, y \geq 0$ Suppose that $x = 0$ and $y \geq 0$. Therefore, we can say

$$|0 \cdot y| = |0| \cdot |y|$$

$$|0| = |0| \cdot |y|$$

$$|0| = |0|$$

Without the loss of generality, we may say that this applies if x or y is 0.

Case 2: $x > 0$ and $y > 0$ Suppose that $x > 0$ and $y > 0$. By **Definition 8.4** we may say that $|x| = x$ and $|y| = y$. As $xy > 0$ then $|xy| = xy$. Therefore, we may say that

$$|xy| = |x| \cdot |y|$$

As $|x| \cdot |y| = x \cdot y$ and $|xy| = xy$ we have that

$$xy = x \cdot y$$

$$xy = xy$$

Case 3: $x < 0$ and $y > 0$. Suppose that $x < 0$ and $y > 0$. Then, we may say that $|x| = -x$, $|y| = y$, and $|xy| = -xy$. Therefore, we may say that

$$|xy| = |x| \cdot |y|$$

As $|x| \cdot |y| = -x \cdot y$ and $|xy| = -xy$ we have that

$$-xy = -x \cdot y$$

$$-xy = -xy$$

Case 4: $x < 0$ and $y < 0$. Suppose that $x < 0$ and $y < 0$. Then, we may say that $|x| = -x$, $|y| = -y$, and $|xy| = xy$. Therefore, we may say that

$$|xy| = |x| \cdot |y|$$

As $|x| \cdot |y| = -x \cdot -y$ and $|xy| = xy$ we have that

$$xy = -x \cdot -y$$

$$xy = xy$$

Without the loss of generality, we may say these cases are exhaustive and work to prove that $|xy| = |x| \cdot |y|$. $\square$

Exercise 8.8, Part (b). We want to show that $x \leq |x|$ and $-x \leq |x|$. We will proceed with three cases:

Case 1: $x > 0$ Suppose that $x > 0$. Therefore, by **Definition 8.4**, we may say that $|x| = x$. Therefore, we can say that $x \leq x$ as $x = x$. Then, as we know that $-x < x$, we may say $-x \leq x$.

Case 2: $x < 0$ Suppose that $x < 0$. Therefore, by **Definition 8.4**, we may say that $|x| = -x$. As $x < 0$, we may say that $-x > x$. Therefore, we may say that $x < -x$ and therefore $x \leq |x|$. Then, as $-x = -x$, we may say that $-x < -x$ and therefore $-x \leq |x|$.

Case 3: $x = 0$ Suppose that $x = 0$. Therefore $|x| = 0$. Hence, we can say that $0 \leq 0$ and $0 \geq 0$. Thus, we can write $x \leq |x|$ and $x \geq |x|$. Therefore, we have that $x \leq |x|$ and $-x \leq |x|$.

These cases suffice to prove that $x \leq |x|$ and $-x \leq |x|$. $\square$

The following lemma is very useful in proofs involving inequalities.

Lemma 8.9. *For any real numbers x , y , and z , we have*

$$(a) \quad |x + y| \leq |x| + |y|,$$

$$(b) \quad |x - z| \leq |x - y| + |y - z|, \text{ and}$$

$$(c) \quad ||x| - |y|| \leq |x - y|.$$

(Hint for (b) and (c): Use (a).)

Proof of Lemma 8.9. We proceed through each part of the lemma using the properties of ordered fields.

Part (a) We want to show that $|x + y| \leq |x| + |y|$. From Exercise 8.8(b), we know that:

$$-|x| \leq x \leq |x| \quad \text{and} \quad -|y| \leq y \leq |y|.$$

By the additive property of ordered fields, we may add these two inequalities together to obtain:

$$-(|x| + |y|) \leq x + y \leq |x| + |y|.$$

By the definition of absolute value, if $-c \leq a \leq c$, then $|a| \leq c$. Taking $a = x + y$ and $c = |x| + |y|$, we conclude that:

$$|x + y| \leq |x| + |y|.$$

Part (b) We want to show that $|x - z| \leq |x - y| + |y - z|$. To use our result from Part (a), we must express $(x - z)$ as a sum of two terms. We do this by adding and subtracting y inside the absolute value:

$$|x - z| = |x - y + y - z| = |(x - y) + (y - z)|.$$

Now, let $a = (x - y)$ and $b = (y - z)$. By applying Part (a) we get:

$$|(x - y) + (y - z)| \leq |x - y| + |y - z|.$$

Combining the equality and the inequality, we have:

$$|x - z| \leq |x - y| + |y - z|.$$

This works to prove that the distance between any two points is less than or equal to the sum of the distances through an intermediate point.

Part (c) We want to show that $||x| - |y|| \leq |x - y|$. First, we write x as $x = (x - y) + y$. Applying Part (a) allows us to write:

$$|x| = |(x - y) + y| \leq |x - y| + |y|.$$

Subtracting $|y|$ from both sides yields $|x| - |y| \leq |x - y|$.

Then, by symmetric reasoning, we write y as $y = (y - x) + x$. Then, from Part (a), we get:

$$|y| = |(y - x) + x| \leq |y - x| + |x|.$$

Subtracting $|x|$ from both sides yields $|y| - |x| \leq |y - x|$. Since $|y - x| = |x - y|$, we have:

$$-(|x| - |y|) \leq |x - y|.$$

Combining the two results, we have $-|x - y| \leq |x| - |y| \leq |x - y|$, which implies $||x| - |y|| \leq |x - y|$. $\square$

Mathematicians often use the letters δ (delta) and ϵ (epsilon) to denote small positive numbers.

Exercise 8.10. Let $a, \delta \in \mathbb{R}$ with $\delta > 0$. Prove that

$$(a - \delta, a + \delta) = \{x \in \mathbb{R} \mid |x - a| < \delta\}.$$

Exercise 8.10. Let $a, \delta \in \mathbb{R}$ where $\delta > 0$. We want to prove that $(a - \delta, a + \delta) = \{x \in \mathbb{R} \mid |x - a| < \delta\}$.

($\Rightarrow$) Let there be an arbitrary $p \in \mathbb{R}$ such that $p \in (a - \delta, a + \delta)$. We want to show that $p \in \{x \in \mathbb{R} \mid |x - a| < \delta\}$. As $p \in (a - \delta, a + \delta)$, we may say that

$$a - \delta < p < a + \delta$$

We can subset a from both sides to get the inequality

$$-\delta < p - a < \delta$$

By **Definition 8.4**, we may say that

$$|p - a| < \delta$$

Therefore, we may say that for some $p \in \mathbb{R}$, $|p - a| < \delta$.

($\Leftarrow$) Let $|p - a| < \delta$. We may write this as $-\delta < p - a < \delta$. Therefore, we may say that $p - a$ lies in a region between $-\delta$ and δ . Then, we may add a on both sides to get $-\delta + a < p < \delta + a$. We get that p lies between $a - \delta$ and $a + \delta$. Therefore, we may say that $p \in (a - \delta, a + \delta)$.

This works to prove that $(a - \delta, a + \delta) = \{x \in \mathbb{R} \mid |x - a| < \delta\}$. □

Lemma 8.11. Let I be an open set containing the point $p \in \mathbb{R}$. Then

- i) there exists a number $\delta > 0$ such that $(p - \delta, p + \delta) \subset I$
- ii) there exists a natural number N such that for all natural numbers $k \geq N$ we have $(p - \frac{1}{k}, p + \frac{1}{k}) \subset I$.

Proof of Lemma 8.11, Part 1. Let I be an open set containing point $p \in \mathbb{R}$. We want to show that there exists $\delta > 0$ such that $(p - \delta, p + \delta) \subset I$. For this, it suffices to show that there exists some case such that the region $(p - \delta, p + \delta)$ lies within the boundaries of I .

Consider I to be defined as the open for (a, b) where $a, b \in \mathbb{R}$. Let $I = \{x \in \mathbb{R} \mid a < x < b\}$. As $I \ni p$, we have have that $p \in \{x \in \mathbb{R} \mid a < x < b\}$ and therefore $a < p < b$. As I is continuous and open, we may say there exists a region around p such that $\exists \delta > 0$ where $a < p - \delta < p + \delta < b$. AS I is open and does not contain a and b , we have that there must be some arbitrary z such that $a < z < p - \delta$ or $p + \delta < z < b$. Therefore, there exists some $\delta > 0$ such that $(p - \delta, p + \delta) \subset I$.

This works to prove that there exists some $\delta > 0$ such that $(p - \delta, p + \delta) \subset I$. □

Proof of Lemma 8.11, Part 2. Secondly, we want to show that if I is an open interval (a, b) , then for every $p \in I$, there exists some $k \in \mathbb{N}$ such that $(p - \frac{1}{k}, p + \frac{1}{k}) \subset I$.

Let $p \in (a, b)$. By the definition of the interval, we have $a < p < b$. This implies that the distances to the endpoints are positive: $p - a > 0$ and $b - p > 0$.

By the Archimedean Principle, there exists a natural number $k \in \mathbb{N}$ such that:

$$\frac{1}{k} < p - a \quad \text{and} \quad \frac{1}{k} < b - p.$$

We want to show that for this choice of k , the neighborhood $(p - \frac{1}{k}, p + \frac{1}{k})$ is contained within (a, b) . Let z be an arbitrary point in this neighborhood. Then, by definition:

$$p - \frac{1}{k} < z < p + \frac{1}{k}.$$

From our Archimedean choice, we can rearrange the inequalities $\frac{1}{k} < p - a$ and $\frac{1}{k} < b - p$ to see that $a < p - \frac{1}{k}$ and $p + \frac{1}{k} < b$. Therefore, by the Transitive Property of order:

$$a < p - \frac{1}{k} < z < p + \frac{1}{k} < b.$$

This shows that $a < z < b$, which means $z \in (a, b)$. Since z was arbitrary, we have shown that $(p - \frac{1}{k}, p + \frac{1}{k}) \subset I$.

Furthermore, for any $m \geq k$, we know that $\frac{1}{m} \leq \frac{1}{k}$. It follows that the interval $(p - \frac{1}{m}, p + \frac{1}{m})$ is contained within $(p - \frac{1}{k}, p + \frac{1}{k})$, and is thus also a subset of I . $\square$

We now discuss the relationship between connectedness and intervals. First however we need to introduce the subspace topology.

Definition 8.12. Let $A \subset X \subset \mathbb{R}$. We say that A is *open in X* if it is the intersection of X with an open set, and *closed in X* if it is the intersection of X with a closed set. (This is called the subspace topology on X).

Remark 8.13. $A \subset \mathbb{R}$ open, as defined in Script 3, is equivalent to A open in $\mathbb{R}$.

Exercise 8.14. Let $A \subset X \subset \mathbb{R}$. Show that $X \setminus A$ is closed in X if, and only if, A is open in X .

Exercise 8.14. Let $A \subset X \subset \mathbb{R}$. We want to prove that $X \setminus A$ is closed in X if and only if A is open in X .

($\Rightarrow$) Suppose that $X \setminus A$ is a closed set in X . We want to show that A must be an open set in X . Let $X \setminus A = X \cap M$ where M is a closed set. Therefore, we may then write:

$$X \setminus (X \setminus A) = X \setminus (X \cap M)$$

$$A = X \setminus (X \cap M)$$

We may write this as

$$A = X \cap (\mathbb{R} \setminus M)$$

As M is closed in $\mathbb{R}$, and as we know that the complement of a closed set is an open set, we may say that $\mathbb{R} \setminus M$ is an open set. Therefore, we may say that A is the intersection of X with an open set $\mathbb{R} \setminus M$. Hence, this works to prove that A is an open set in X .

($\Leftarrow$) Suppose that A is an open set. Therefore, we may say that there exists some open set N such that $A = X \cap N$.

Now, we may take the complement of both sides of the equation such that

$$X \setminus A = X \setminus (X \cap N)$$

We may write this as

$$X \setminus A = X \cap (\mathbb{R} \setminus N)$$

As N is open, we may say that $\mathbb{R} \setminus N$ is closed. Therefore, we may say that $X \setminus A$ is the intersection of X and a closed set $\mathbb{R} \setminus N$. Hence, this works to show that if A is open in X , then $X \setminus A$ is closed in X .

As we have shown that this applies for both cases, it works to prove that $X \setminus A$ is closed in X if and only if A is open. $\square$

Exercise 8.15. a) Let $[a, b] \subset \mathbb{R}$. Give an example of a set $A \subset [a, b]$ such that A is open in $[a, b]$ but not in $\mathbb{R}$.

b) Give an example of sets $A \subset X \subset \mathbb{R}$ such that A is closed in X but not in $\mathbb{R}$.

Exercise 8.15 (a). Let $[a, b] \subset \mathbb{R}$. We want to provide a case that such that set $A \subset [a, b]$ is open in $[a, b]$ but not in $\mathbb{R}$.

Consider the set $[2, 6] = \{2, 3, 4, 5, 6\}$. Let there be an open set U which is defined as $(3, \infty)$. Let $A = [2, 6] \cap U$. We may write this as

$$A = [2, 6] \cap (3, \infty)$$

$$A = \{2, 3, 4, 5, 6\} \cap (3, \infty)$$

$$A = (3, 6]$$

By **Definition 8.12**, we may say that A is open in $[2, 6]$ as it is defined as the intersection of $[2, 6]$ and an open set. Furthermore, we know that $(3, 6]$ is not open in $\mathbb{R}$ as $\mathbb{R} \setminus (3, 6]$ does not contain

the limit point 6. Therefore, this works as an example of a set that is open in $[a, b]$ but not open in $\mathbb{R}$. $\square$

Exercise 8.15(b). Let $A \subset X \subset \mathbb{R}$. We want to provide an example of a set A such that A is closed in X but not closed in $\mathbb{R}$.

Consider the set $(2, 6)$. Then, let there be a closed set M which is defined as $(-\infty, 5)$. Let $A = (2, 6) \cap (-\infty, 5)$. We may write this as:

$$A = (2, 6) \cap (-\infty, 5)$$

Therefore, we may write set A as

$$A = (2, 5]$$

Which is a subset of X

By **Definition 8.12**, we may say that A is closed in $(2, 6)$ as it is defined as the intersection of $(2, 6)$ and an closed set $(-\infty, 5)$. Furthermore, we may know that A is not closed in $\mathbb{R}$ as it does not contain the limit point 2. Therefore, this works as an example of a set A that is closed in X but not closed in $\mathbb{R}$. $\square$

Now that we have defined the subspace topology, Definition 4.22 tells us that a set $X \subset \mathbb{R}$ is *connected* if it cannot be written as the union $X = A \cup B$ of disjoint, non-empty sets A and B that are open in X .

Theorem 8.16. *Let $X \subset \mathbb{R}$ be nonempty. Then X is connected if, and only if, X is an interval.*

Proof of Theorem 8.16. Let $X \subset \mathbb{R}$ be nonempty. We want to prove that X is connected if and only if X is an interval.

($\Rightarrow$) Firstly, let X be connected. We want to show that X is an interval. For this, it suffices to show that X takes the form (a, b) .

For the sake of contradiction, suppose that X is connected and we cannot write it in the form of an interval. Then there exist points $x, y \in X$ with $x < y$ and a real number z such that

$$x < z < y \quad \text{and} \quad z \notin X.$$

Therefore, as $X = X \cup B$ and $z \notin X$ we may define the sets A and B as

$$A := X \cap (-\infty, z) \quad \text{and} \quad B := X \cap (z, \infty).$$

Both A and B are nonempty, and clearly $A \cap B = \emptyset$ since no element can be simultaneously less than and greater than z . Moreover, each of the sets A and B is open in the subspace topology on

X , as they are intersections of X with open intervals in $\mathbb{R}$.

By expressing X as the union of two non-empty, disjoint, relatively open subsets, we have shown that X is disconnected. This contradicts our initial assumption that X is connected. Therefore, no such point z can exist outside of X if it lies between two points of X . This confirms that X must be an interval.

($\Leftarrow$) Secondly, let X be an interval. We want to show that X must be connected. For this it suffices to show when $X = A \cup B$, $A \cap B \neq \emptyset$. Since A and B are non-empty, we can pick a point $a \in A$ and a point $b \in B$. Without loss of generality, assume $a < b$. Because X is an interval and $a, b \in X$, the entire closed interval $[a, b]$ must be contained in X .

Now, consider the set $M = \{x \in [a, b] \mid [a, x] \subseteq A\}$. This set M is non-empty (because $a \in M$) and it is bounded above by b . Therefore, M must have a least upper bound. Let $m = \sup(M)$.

Let $x \in X$ where X takes the form (a, b) . Therefore, we may say that $a < x < b$. If $X = A \cup B$, then we may say that $x \in A$ or $x \in B$. Now, we want to show that $A \cap B \neq \emptyset$. Consider two cases:

Case 1: $m \in A$ If $m \in A$, then since A is open in X , there exists $\epsilon > 0$ such that $(m - \epsilon, m + \epsilon) \cap X \subset A$. This implies there are points in A greater than m , contradicting that m is an upper bound of M .

Case 2: $m \in B$ If $m \in B$, then since B is open in X , there exists $\epsilon > 0$ such that $(m - \epsilon, m + \epsilon) \cap X \subset B$. This implies no points of A can exist in $(m - \epsilon, m]$, contradicting that m is the least upper bound of M .

Thus, m can neither be in A nor B , contradicting $X = A \cup B$. Therefore, X is connected. $\square$

Finally, we briefly explore functions on intervals.

Definition 8.17. Let I be an interval and let $f : I \rightarrow \mathbb{R}$.

- a) We say that f is *increasing* on I if, whenever $x, y \in I$, with $x < y$, $f(x) \leq f(y)$.
- b) We say that f is *decreasing* on I if, whenever $x, y \in I$, with $x < y$, $f(x) \geq f(y)$.
- c) We say that f is *strictly increasing* on I if, whenever $x, y \in I$, with $x < y$, $f(x) < f(y)$.
- d) We say that f is *strictly decreasing* on I if, whenever $x, y \in I$, with $x < y$, $f(x) > f(y)$.

Note that “strictly increasing” means the same thing as “order preserving.”

Lemma 8.18. If f is strictly increasing or strictly decreasing on an interval I then f is injective on I .

Proof of Lemma 8.18. We want to prove that if f is strictly increasing or strictly decreasing on an interval I , then f is injective on I . We will prove this by contradiction.

Suppose f is not injective on I . By the definition of injectivity, this implies there exist distinct elements $x, y \in I$ such that $f(x) = f(y)$ even though $x \neq y$. By the Law of Trichotomy, since $x \neq y$, we must have either $x < y$ or $y < x$. Without loss of generality, let $x < y$.

Suppose f is strictly increasing on I . Then, by **Definition 8.17**, the condition $x < y$ implies that $f(x) < f(y)$. However, this contradicts our assumption that $f(x) = f(y)$, as a value cannot be both strictly less than and equal to another value.

Now suppose f is strictly decreasing on I . Then, by **Definition 8.17**, the condition $x < y$ implies that $f(x) > f(y)$. However, this also contradicts our assumption that $f(x) = f(y)$, as a value cannot be both strictly greater than and equal to another value.

In either case, assuming f is not injective leads to a contradiction of the definition of strictly monotonic functions. Therefore, it must be that if f is strictly increasing or strictly decreasing on an interval I , then f is injective on I . $\square$

Additional Exercises

1. Let $X \subset \mathbb{R}$. Prove that X is both open and closed in X .
2. (a) Let $X \subset \mathbb{R}$ be a closed set. Show that $A \subset X$ is closed in X if and only if A is closed in $\mathbb{R}$.
 (b) Let $X \subset \mathbb{R}$ be an open set. Show that $A \subset X$ is open in X if and only if A is open in $\mathbb{R}$.
3. Suppose that $A, B \subset \mathbb{R}$ are connected.
 - (a) Show that $A \cap B$ is connected.
 - (b) Give a non-trivial¹ condition on A and B so that $A \cup B$ is connected.

¹I.e. you may not use the condition $A = B$ or the condition that either A or B is empty.